PROFILE

Data and AI leader with over fourteen years in software and data, including more than ten in enterprise data architecture, and more than three years leading technical teams, built inside regulated medical technology organizations at Medtronic and Boston Scientific. Sets data strategy and architecture, governs enterprise data standards, and runs analytics delivery as a service function, most recently operationalizing enterprise data governance for AI/ML workflows and agentic AI pipelines. Works to HIPAA, IRB, and FDA regulatory requirements for data privacy and compliance. Track record of building integrated data ecosystems that serve large research and business user populations, including a platform serving over 1,500 users that delivered $2.4M in annual savings, and of translating technical architecture decisions for executive audiences.

SKILLS

Data Strategy, Architecture, and Governance:

Enterprise data architecture, data strategy and roadmap, data governance and standards, data quality frameworks, common data modeling, enterprise data interoperability, Data Mesh, Medallion Architecture, enterprise data warehouse (EDW) design, semantic modeling, data integration, data stewardship, data management practices

AI, Machine Learning, and Data Science:

Machine learning operations (MLOps), model deployment and monitoring, feature stores, agentic AI pipelines, large language model (LLM) solutions, cloud-native AI scaling, advanced analytics, computational methods, medical imaging computer vision, deep learning, statistical analysis

Platforms and Languages:

Cloud platforms: AWS, Azure. Languages: Python, R, SQL (SQL Server, Oracle, PostgreSQL). Data platforms: Databricks, Snowflake, Delta Lake, Redshift, EMR. Pipelines and orchestration: ETL/ELT, dbt, Apache Airflow, Alteryx, Spark. Business intelligence: Tableau, Power BI, Amazon QuickSight. Containers: Kubernetes, Docker.

Regulatory, Privacy, and Compliance:

Data privacy and compliance, HIPAA, Institutional Review Board (IRB) requirements, FDA-regulated environments, audit-ready data assets, access control and platform security, regulated data handling

Leadership and Delivery:

Team leadership and mentoring, staffing and performance management, portfolio and stakeholder management, service delivery models, prioritization and resource allocation, executive communication, cross-functional partnership

EXPERIENCE

Senior Data Engineer, Data and AI | Medtronic, Minneapolis, MN — Jan 2026 to Present

- Led a cross-functional team building domain-oriented data products on cloud platforms (AWS, Azure), enforcing Medallion Architecture and Data Mesh principles to make data assets reusable and scalable across the enterprise.
- Operationalized enterprise-wide data governance standards for AI/ML workflows, agents, and skills, ensuring compliance and platform security in a regulated medical device environment.
- Partnered with business stakeholders to deliver streamlined data products and advanced analytics, improving decision-making velocity by 20% within three months of joining.
- Architected scalable AI and MLOps pipelines and implemented feature stores, establishing repeatable infrastructure for model delivery.
- Defined technical strategy and mentored team members on agentic AI pipelines, optimizing production performance.
- Collaborated with product teams to define the roadmap and technical strategy for cloud-native AI initiatives.
- Core stack: cloud platforms (AWS, Azure), SQL, Python, and enterprise BI.

Senior Data Engineer, Data and AI | Boston Scientific, Arden Hills, MN — Feb 2024 to Dec 2025

- Owned an enterprise data and analytics platform serving over 1,500 users across North America, EMEA, APAC, and LATAM, delivering $2.4M in annual savings through platform consolidation and process improvement.
- Implemented Medallion architecture (Bronze, Silver, Gold) on cloud platforms to achieve enterprise scalability and consistent data quality tiers.
- Designed KPI frameworks, semantic models, and executive reporting across HR, Finance, and IT data, and modernized the analytics estate through cloud migration, automation, and governed reporting.
- Core stack: cloud platforms (AWS, Azure), SQL, Python, and enterprise BI.
- Reduced the analytics project backlog by 40% within 12 months by coaching and building a high-performing cross-functional team, improving intake and delivery throughput for competing analytic requests.

- Built employee attrition forecasting models using gradient boosting and survival analysis, and ARIMA-based headcount projection models supporting the Annual Operating Plan, converting workforce reporting from a backward view into a forward one for HR and Finance leadership.
- Increased user adoption by 10% through a strategic migration to centralized analytics interfaces.

Senior Data Engineer, Consultant, Cardiology Sales Operations | Boston Scientific, Maple Grove, MN Nov 2022 to Feb 2024

- Accelerated insight delivery by 40% within six months for Cardiology Sales Operations (CRMDx, AFS, ICTx) by implementing enterprise-grade analytics platforms.
- Developed automated Alteryx pipelines and Tableau dashboards, producing audit-ready data assets aligned to regulated business requirements.

Senior Data Engineer | Crowd Capital, Miami, FL (remote) Mar 2022 to Oct 2022

- Launched an Automated Property Scoring Module on AWS and Databricks that scaled decision-making from 20 to over 1,000 bids within six months, increasing quarterly recurring revenue by $300K.
- Engineered big data pipelines in Databricks and optimized distributed compute workloads for housing analytics.

Lead Data Engineer | Global Overview, Minneapolis, MN Oct 2020 to Feb 2022

- Delivered over $6M in cost savings and revenue enablement by mentoring a team of six and streamlining retail, operations, and marketing analytics across 100+ vendor accounts and 150+ advertising brands.
- Built marketing spend and multi-touch attribution models across display, over-the-top, and search channels, using A/B testing to optimize cross-channel allocation.
- Reduced data warehouse costs by $2M annually by consolidating over 560 legacy tables and improving reporting performance from multi-minute waits to two-second refreshes.

Data Scientist | Cytilife, Minneapolis, MN Feb 2019 to Oct 2020

- Built analytical infrastructure on AWS (Redshift, EMR, Athena, Glue, Kinesis, Lambda) connecting structured and unstructured sources, processing datasets from 100,000 to millions of rows.
- Developed and deployed a machine learning forecasting model for resource utilization, hosted as a REST API on Amazon SageMaker for consumption by mobile applications.
- Delivered a real-time computer vision people-counting application and dashboard for university administration using OpenCV and Python.
- Transformed raw data into actionable business insights using Amazon QuickSight for executive decision-making.

Research Assistant | University of St. Thomas, Minneapolis-St. Paul, MN Mar 2018 to Dec 2020

- Conducted applied machine learning research on National Science Foundation funding proposal abstracts, spanning 34 years and hundreds of thousands of records, producing document embeddings and unsupervised clusters to support the merit review process.
- Authored the resulting paper, selected for SAS Global Forum 2020, with the team placing first in the nationwide SAS competition.
- Supported faculty and student research projects across big data and software development, translating research questions into data pipelines and reproducible analysis.
- Deployed a research model as a REST API on Amazon SageMaker using Docker and Elastic Container Registry for the AWS Marketplace competition.
- Placed in national data science competitions including MinneMUDAC and FastCon 2019 (multivariate LSTM soybean futures forecasting, second place) and the AIM Consulting Hackathon (imbalanced-data classification, first place).

Senior Build Analyst | Amphora, Hyderabad, India Feb 2012 to May 2017

- Built a trade aggregation service that ingested bulk trade data from global commodity exchanges into the firm's trading and risk management platform, removing manual preprocessing for downstream users.
- Developed the enterprise notifications service for email, push, and in-app delivery, using A/B testing to optimize send timing and building the batch processing and queuing behind it.
- Led the development team's migration from Team Foundation Server to Git and from waterfall to SCRUM, delivering the transition without disrupting active development.
- Automated build, test, and release pipelines, and built a tool for controlling application services across multiple remote server environments.

RESEARCH, PUBLICATIONS, AND APPLIED MACHINE LEARNING

Publication: Unsupervised Contextual Clustering of Abstracts, SAS Global Forum 2020. Applied Doc2Vec, BERT, and K-means clustering to National Science Foundation funding proposal abstracts to recommend similar proposals for merit review. First place, nationwide SAS competition.

Medical Imaging: Diabetic retinopathy detection from retinal fundus images. Deep learning image classification using Fastai, PyTorch, and OpenCV. Ranked in the top 11% (298 of 2,943 participants).

Natural Language Processing: Toxicity classification with gender-bias mitigation using BERT and LSTM architectures. Ranked in the top 8% (225 of 3,165 participants).

Time Series Forecasting: Multivariate LSTM model predicting soybean futures contract prices, MinneMUDAC 2019. Team placed second on 15-day forward predictions.

Classification: Machine learning classifier on a highly imbalanced dataset, AIM Consulting Hackathon, first place. Crowdfunding outcome prediction using RandomForest, XGBoost, and LightGBM.

Computer Vision: Semantic image segmentation with U-Net architecture in PyTorch for per-pixel object classification.

EDUCATION

M.S., Software Management | University of St. Thomas, MN | GPA 3.89

M.S., Data Science | University of St. Thomas, MN | GPA 3.85

B.S., Computer Science | Sardar Vallabhbhai National Institute of Technology, India | GPA 3.30